

RKDF University, Bhopal
Faculty Profile

Basic Information				
Name	Dr. Akant Kumar Raghuwanshi			
Date of Birth	25.05.1986			
Designation	Associate Professor			
Department	Electronics & Communication Engineering			
Experience	13 year			
Email ID	akantthakur7@gmail.com			
Contact No	9752825636			
Educational Qualifications				
Description	Year	%	Institute/University	
(UG)-	2009	69.19	RGPV	
(PG)-	2014	7.08	RGPV	
M. Phil.	NA	NA		
(Ph. D.)-	Jan-2023	Awarded	RKDF University	
Post Doctorate	NA	NA		
NET Qualified/GATE				
Experience Detail				
Experience (Teaching/Research)	Designation	Duration		Name of Institute/University
		From	To	
Teaching	Lecturer	21.05.2012	30.06.2014	Vedica Institute of Technology
Teaching	Asst. Professor	01.07.2014	05.05.2025	Vedica Institute of Technology
Publications				
No. of Papers Published (Attach Annexure of papers published in peer-reviewed or refereed journals)		32		
No. of Books Published		NA		
Books Chapters Published (Provide print of		NA		

1st Page of Book			
No. of Patents Published/Grant	02		
Ph. D/M. Phil Project supervised			
Research Program	Award	Under Supervision	Name of University
Ph. D (Provide detail i.e. name, title etc)	NA	NA	NA
M. Phil	NA	NA	NA
PG Thesis/Dissertation	10	1	RKDF University
Area of Expertise (100 words)			
Expertise in developing and optimizing power-aware routing protocols to enhance the lifetime and energy efficiency of Wireless Sensor Networks (WSNs). Focus on minimizing energy consumption while maintaining reliable communication, using techniques like adaptive routing, energy-aware algorithms, and dynamic power control. Expertise includes evaluating trade-offs between energy and performance, ensuring optimal sensor deployment, and designing protocols that address energy constraints to extend network longevity and improve overall performance in resource-limited environments. Applying genetic algorithms to optimize digital circuit performance, focusing on parameter tuning, design refinement, and resource management to maximize efficiency, speed, and power consumption while ensuring reliability.			
Award and Achievement			
Name of Award	Description (With certified of Copy award)		
National	NA		
International	NA		
Conference/Seminar/Workshops/FDP			
Description	No.		
Conference/Seminar p a p e r presentation	6		
Conference/Seminar attended/ organized	NA		
Work shop attended/ organized	NA		
FDP Attended/ organized	NA		
Research Project			
Name of Project	Funding Agencies	Amount	

- **Any other Achievement**

List of Publications	Hyper Link
“Relative Entropy Based Energy Efficient Multi-Hop Forwarding Routing Protocol for WSN” Accepted and will be published in Samriddhi Journal, Volume 14(4), 2022.	https://www.smsjournals.com/index.php/SAMRIDDHI/article/view/2989
“Energy Efficient Access Control MAC Layer Protocol over Wireless Sensor Network” Samriddhi Journal, Volume 14(3), 2022, pp. 270-279	https://www.smsjournals.com/index.php/SAMRIDDHI/search/search

- **List of Patent :**

1. Patent design numbered **202321064948**
Title “Wireless power transfer strategy for implantable biomedical devices”.
2. Patent design numbered **6435770**
Title “AI Based Portable Wireless Communication Device”.

PUBLICATIONS/CONFERENCES

1. **Akant Kumar Raghuwanshi**, Ajay Barapatre and Balajee Sharma (2025) Wireless Sensor Networks and Its Clustering Algorithm. Annual International Conference on Recent advanced in Engineering Technology , Healthcare& Management (AIC-RAETHM 2025) *Jan10,11 2025*.
2. Balajee Sharma, **Akant Kumar Raghuwanshi** and Ajay Barapatre (2025) Emerging Trends in 5G and Beyond-5G Wireless Networks. Annual International Conference on Recent advanced in Engineering Technology, Healthcare& Management (AIC-RAETHM 2025) *Jan10,11 2025*.
3. Ajay Barapatre, Balajee Sharma and **Akant Kumar Raghuwanshi** (2025) A Study on Security Requirements of IOT. Annual International Conference on Recent advanced in Engineering Technology , Healthcare& Management (AIC-RAETHM 2025) *Jan10,11 2025*.
4. Vishal Baraiya, **Akant Kumar Raghuwanshi**, “Review of Existing Fingerprint Recognition Techniques and an Enhanced Methodology. “Research Journal of Engineering Technology and Management (ISSN : 2582-0028, Volume 07, Issue 03, June-2024.
5. Sangeeta Niranjana, **Akant Kumar Raghuwanshi**, “An Enhanced Data Security Using RSA Algorithm with Integration of Vedic Multiplier. “Research Journal of Engineering Technology and Management (ISSN : 2582-0028, Volume 07, Issue 03, June-2024.
6. Shivhari Arya, **Akant Kumar Raghuwanshi**, “Review of Recent Advanced Image Encryption Techniques. “Research Journal of Engineering Technology and Management (ISSN : 2582-0028, Volume 07, Issue 03, June-2024.
7. Shivhari Arya, **Akant Kumar Raghuwanshi** “Review of Recent Advanced Image Encryption Techniques. “Shodh Sangam – A RKDF University Journal of Science and Engineering” ISSN No. 2581-5806, Vol-08, No. 04, July-Aug – 2024.

8. Sangeeta Niranjana, **Akant Kumar Raghuwanshi** "An Enhanced Methodology for Data Security Using RSA Algorithm. "Shodh Sangam – A RKDF University Journal of Science and Engineering" ISSN No. 2581-5806, Vol-08, No. 04, July-Aug – 2024.
9. Vijay Baraiya, **Akant Kumar Raghuwanshi** "An Advanced Fingerprint Recognition Using Hybrid Deep Learning. "Shodh Sangam – A RKDF University Journal of Science and Engineering" ISSN No. 2581-5806, Vol-08, No. 04, July-Aug – 2024.
10. Balajee Sharma and **Akant Kumar Raghuwanshi** (2024) Advancements in AI-driven Techniques for MIMO-OFDM Systems: A Comparative Analysis. *International Journal of Innovative Research in Technology and Science*, 11(02).
11. **Akant Kumar Raghuwanshi** and Balajee Sharma (2024) Investigating Power-Aware Protocols for Wireless Sensor. *International Journal of Innovative Research in Technology and Science*, 11(02).
12. **A.K. Raghuwanshi** and V.S. Chaudhary, "Relative Entropy Based Energy Efficient Multi-Hop Forwarding Routing Protocol for WSN" Accepted and will be published in Samriddhi Journal, Volume 14(4), 2022.
13. **A.K. Raghuwanshi** and V.S. Chaudhary, "Energy Efficient Access Control MAC Layer Protocol over Wireless Sensor Network" Samriddhi Journal, Volume 14(3), 2022, pp. 270-279
14. **A.K. Raghuwanshi** and V.S. Chaudhary, "Comparative analysis of Energy Efficient Routing Protocol over WSN," International Conference on Computational Intelligence and computing applications (ICCICA), 2021, pp. 1-6 (Scopus Index)
15. **A.K. Raghuwanshi** and V.S. Chaudhary, "Comparative analysis of Energy Efficient Routing Protocol over WSN," International Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, 2018, pp. 38-41.
16. Patel Hirenkumar Sureshbhai, **Akant Kumar Raghuwanshi** "Research Journal of Engineering Technology and Medical Sciences (ISSN : 6212), Volume 05, Issue 02, June 2022.
17. Patel Hirenkumar Sureshbhai, **Akant Kumar Raghuwanshi** "Methods of Cognitive Spectrum Sensing in DWT based MIMO-OFDM: A Review. "Shodh Sangam – A RKDF University Journal of Science and Engineering" ISSN No. 2581-5806, Vol-05, No. 04, August – 2022.
18. Rupali Bende, **Akant Kumar Raghuwanshi**, "Deep Learning Based Compressed Sensing in WBAN, "Research Journal of Engineering Technology and Medical Sciences (ISSN: 2582-6212), Volume 04, Issue 03, September-2021.
19. Rupali Bende, Akant Kumar Raghuwanshi, "Compressed Sensing based on Deep Learning in Wireless Body Networks "Shodh Sangam – A RKDF University Journal of Science and Engineering"
20. Pradeep Kumar Suryawanshi, Akant Kumar Raghuwanshi, "Cyclostationary Spectrum Sensing in Cognitive Radio : A Review. "Shodh Sangam – A RKDF University Journal of Science and Engineering" ISSN No. 2581-5806, Vol-03, No. 01, Jan - 2020.
21. Anil Kumar Malviya, Akant Kumar Raghuwanshi, "Compressed Sensing Space Time Block Code OFDM System for Optical Communication. "Research Journal of Engineering Technology and Management (ISSN : 2582-0028, Volume 03, Issue 02, June-2020.
22. Anil Kumar Malviya, Akant Kumar Raghuwanshi, "A Novel Orthogonal Frequency Division Multiplexing (OFDM) System for Optical Communication "Shodh Sangam – A RKDF University Journal of Science and Engineering" ISSN No. 2581-5806, Vol-03, No. 04, July - 2020.

23. Pradeep Kumar Suryawanshi, Akant Kumar Raghuwanshi, "Cyclostationary Spectrum Sensing using AIC Block in Cognitive Radio "Research Journal of Engineering Technology and Management (ISSN : 2582-0028), Special Issue I (2019)
24. Vishnu Dutt Gautam, Prof Akang Kumar Raghuwanshi, "Underwater Acoustic Communication using multiple input multiple out orthogonal frequency division multiplexing. ISSN (Online) : 2278-3814, Volume8, Issue-5, May, 2019.
25. Vishnu Datt Gautam, Mr. Akant Kumar Raghuwanshi "A Comparative Analysis of Different UWAC Techniques" "Shodh Sangam – A RKDF University Journal of Science and Engineering"
26. Raksha Sharma, Akant Kumar Raghuwanshi, "A Review over Digital Image Processing with Enhancement Approaches, Engineering Universe for Scientific Research ana Management (International Journal) Vol. 7 Issue 2 February 2017.
27. Raksha Sharma, Akant Kumar Raghuwanshi, "An Approach to Digitize Historical Document Using DWT. "Engineering Universe for Scientific Research and Management (International Journal) Vol. 7 Issue 4 April 2017.
28. Nitesh Bathma, Akant Kumar Rathuwanshi, "A Critical Review on Voice Based Recognition System using Computationally Effiient Algorithms". International Journal of Innovative Trends in Engineering (IJITE) ISSN : 2395-2946 Issue : 50, Volume 30, Number 01, 2017.
29. Nitesh Bathma, Akant Kumar Rathuwanshi, "A Voice Based Recognition System Uisng Vector Quantization And Mel Frequency Cepstral Coefficients", International Journal of Scientific Progress and Research (IJSPR) ISSN : 2349-4689 Issue 105, Volume 36, Number 03, 2017.
30. Trapti Saxena, Akant Kumar Raghuwanshi, "International Journal of Engineering Sciences & Management", "Dewalink Precoding Methods for Multi-user Multiple Input-Multiple Output (MU-MIMO) ISSN : 2277-5528, 6(3) : July-Sep, 2016.
31. Trapti Saxena, Akant Kumar Raghuwanshi, "International Journal of Engineering Sciences & Management", "A Review of Downlink Precoding Techniques and Methods for MIMO" ISSN : 2277-5528, 6(3) : July-Sep, 2016.
32. Akant Kumar Raghuwanshi, Sandeep Shrivastva, "A Review on Genetic Aproach over Digital Circuit" International Journal of Computational Intelligence and Information Security, September 2013 Vol. 4 No. 7

• List of Conference:

33. **Akant Kumar Raghuwanshi**, Ajay Barapatre and Balajee Sharma (2025) Wireless Sensor Networks and Its Clustering Algorithm. Annual International Conference on Recent advanced in Engineering Technology , Healthcare & Management (AIC-RAETHM 2025) *Jan10,11 2025*.
34. Balajee Sharma, **Akant Kumar Raghuwanshi** and Ajay Barapatre (2025) Emerging Trends in 5G and Beyond-5G Wireless Networks. Annual International Conference on Recent advanced in Engineering Technology, Healthcare& Management (AIC-RAETHM 2025) *Jan10,11 2025*.
35. Ajay Barapatre, Balajee Sharma and **Akant Kumar Raghuwanshi** (2025) A Study on Security Requirements of IOT. Annual International Conference on Recent advanced in Engineering Technology , Healthcare& Management (AIC-RAETHM 2025) *Jan10,11 2025*.

36. **A.K. Raghuwanshi** and V.S. Chaudhary, "Comparative analysis of Energy Efficient Routing Protocol over WSN," International Conference on Computational Intelligence and computing applications (ICCICA), 2021, pp. 1-6 (Scopus Index)
37. **A.K. Raghuwanshi** and V.S. Chaudhary, "Comparative analysis of Energy Efficient Routing Protocol over WSN," International Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, 2018, pp. 38-41.